

Md. Taifur Hossain.

CS UNDERGRADUATE · 3D VISION & ML SYSTEMS

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Final-year Computer Science undergraduate focused on memory-efficient 3D Gaussian Splatting. Released four open-source tools and opened two pull requests to gsplat, each taken from experiment to a tested, documented result.

EDUCATION

International University of Business Agriculture & Technology (IUBAT) Dhaka, BD
B.Sc. in Computer Science & Engineering (BCSE) · CGPA 3.3 / 4.00 Expected 2027

PUBLICATIONS

Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

Designed a training-free post-processing pipeline (adaptive pruning → outlier cleaning → spherical-harmonic reduction → mixed-precision quantization) achieving a **74% mean size reduction** across three scenes at a 1.3 to 2.7 dB PSNR cost. Diagnosed a per-column INT8 quantization collapse (~15 dB uniform) and recovered to within **0.1 to 0.7 dB of FP16** using Morton Z-order sub-group ranges, a positive and reproducible quantization finding. Isolated a zero-update failure in post-pruning fine-tuning (stale optimizer state), reported as a negative result. Ran the full multi-scene pipeline on a single 16 GB Kaggle T4 GPU using the Nerfstudio / gsplat ecosystem.

OPEN SOURCE CONTRIBUTIONS

gsplat PR #1063 (open): opacity × area weighted K-means for the PngCompression SH codebook, **+0.111 dB** mean PSNR on 9 MipNeRF360 scenes and +0.052 dB on Tanks & Temples, better on every scene at the same size and format.

gsplat PR #1061 (open): fixed PngCompression failing on sh_degree=0 scenes and empty tensors.

PROJECTS

splat-slim Python · Typer · NumPy · CI · repo

Training-free CLI / library that compresses trained 3DGS .ply files ~74% with <3 dB PSNR loss; ships self-decodable Morton-ordered sub-group INT8 with per-group ranges in a sidecar .meta.npz.

splat-lab Vite · TypeScript · Three.js · live demo

Real-time in-browser demo of the pipeline with live sliders and an honest size readout; a TypeScript port of splat-slim.

nerf-prep Python · Docker · GitHub Actions · repo

One-command Dockerized COLMAP → Nerfstudio prep: capture-aware matching, GPU/CPU auto-detect, registration report.

paper-chat Python · Streamlit · sentence-transformers · repo

RAG over a PDF / arXiv library with per-(paper, page) citations; declines when its sources don't cover the question.

lumina-lms Next.js · React · TypeScript · Prisma · repo

Full-stack AI learning management system: a retrieval-augmented tutor that answers from course materials and cites its sources, role-based access for students, instructors and admins, adaptive learning paths, gamification and analytics dashboards.

medical-expert-system Python · Flask · scikit-learn · repo

Diagnosis system ensembling rule-based confidence factors, A* search, Bayesian inference and a Decision Tree + MLP.

TECHNICAL SKILLS

LANGUAGES Python, C, C++, C#, TypeScript / JavaScript, Bash, LaTeX

ML & 3D VISION 3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, PyTorch, scikit-learn, COLMAP, model quantization, Bayesian inference, RAG & embeddings

TOOLS & INFRA Next.js, React, Prisma, Flask, Streamlit, Three.js, Vite, Docker, Git, GitHub Actions (CI/CD), Linux

REFERENCES

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